

Problem Set 4

In this problem set, you will continue to be asked to work with the gamma distribution and you may find it useful to refer back to the previous problem set and/or solutions. I recommend Wikipedia as a reference, https://en.wikipedia.org/wiki/Gamma_distribution.

Keep your rendered `.pdf` to no more than 6 pages. Only provide code in the rendered `.pdf` when it is specifically requested.

1. Continue using four sets of values for the shape parameter α and rate parameter λ : $(\alpha_1, \lambda_1) = (0.5, 0.5)$, $(\alpha_2, \lambda_2) = (2, 1)$, $(\alpha_3, \lambda_3) = (0.5, 1)$, and $(\alpha_4, \lambda_4) = (2, 0.5)$, and store these values in a matrix `Y` with 4 rows and 2 columns, where the first column corresponds to the shape parameter, the second column corresponds to the rate parameter, and the rows correspond to the parameters in the order provided above.
 - (a) Create a matrix `W` that with 4 rows and 100,000 columns. Fill each row of `W` with 100,000 draws from the corresponding gamma distribution using the `rgamma` function. Print the code you use to create this matrix to the rendered `.pdf`.
 - (b) Create a matrix `V` that with 4 rows and 100,000 columns. Fill each row of `V` with 100,000 draws from the corresponding gamma distribution using the `rgamma` function obtained by simulating from gamma distributions with shape 0.5 or 2 and rate 1 using `rgamma` and applying simple mathematical operations. Print the code you use to create this matrix to the rendered `.pdf`. You may need to remind yourself of some gamma distribution facts.
 - (c) Again, consider a sequence of values given by `x <- seq(0.01, 6, by = 0.01)`. Create two matrices `pW` and `pV` each with the same number of rows as `W` and the same number of columns as the length of `x` minus 1 with all elements equal to `NA`. Fill each element of `pW` and `pV` with the proportion of elements in the corresponding row of `W` or `V` between the corresponding element of `x` and the next largest element of `x`. Print the code you use to create these matrices matrix to the rendered `.pdf`.
 - (d) Create a vector `xmid` that has the same number of elements as the length of `x` minus one, and set the elements of `xmid` to be the midpoints between each element of `x`. Print the code you use to create this matrix to the rendered `.pdf`.

- (e) Create a single plot that includes 8 lines corresponding to each row of $\mathbf{pW}$ and each row of $\mathbf{pV}$ plotted against $\mathbf{xmid}$. Use different colors lines to reflect the parameters of the gamma distribution each row of $\mathbf{pV}$ or $\mathbf{pW}$ describes and use different line types for whether or not the line was computed from $\mathbf{pV}$ or $\mathbf{pW}$. Make sure that your plot is clearly annotated and has an informative legend.
 - (f) Based on what you find in (e), explain in at most one sentence whether or not the approaches for simulating gamma random variables in (a) and (b) are equivalent.
 - (g) Create two matrices $\mathbf{qW}$ and $\mathbf{qV}$ each with the same number of rows as $\mathbf{W}$ and the same number of columns as the length of $\mathbf{x}$ with all elements equal to `NA`. Fill each element of $\mathbf{qW}$ and $\mathbf{qV}$ with the proportion of elements in the corresponding row of $\mathbf{W}$ or $\mathbf{V}$ that are less than or equal to the corresponding element of $\mathbf{x}$. Create a third matrix $\mathbf{qT}$ with the same dimensions as $\mathbf{qW}$ and $\mathbf{qV}$. Fill each element of $\mathbf{qT}$ with the probability that a gamma random variable with parameters corresponding to the same row of $\mathbf{Y}$ is less than or equal to the corresponding element of $\mathbf{x}$. Use one of the functions that starts with a letter and ends with `gamma`. Print the code you use to create $\mathbf{qT}$ to the rendered `.pdf`.
 - (h) Create a single plot that includes 12 lines corresponding to each row of $\mathbf{qW}$, each row of $\mathbf{qV}$, and each row of $\mathbf{qT}$ plotted against $\mathbf{x}$. Use different colors lines to reflect the parameters of the gamma distribution each row of $\mathbf{qV}$, $\mathbf{qW}$, or $\mathbf{qT}$ describes and use different line types for whether or not the line was computed from $\mathbf{qV}$, $\mathbf{qW}$, or $\mathbf{qT}$. Make sure that your plot is clearly annotated and has an informative legend.
 - (i) In at most two sentences, explain the differences between lines corresponding to the same gamma distribution in (h), if there are any.
2. This question is introducing the multivariate normal distribution, which I don't expect you to be familiar with yet, and the computational challenges that come up when we work with it.
- (a) Simulate 3 independent standard normal variables and store them in a vector called $\mathbf{z}$. Print the code you use to create $\mathbf{z}$ to the rendered `.pdf`.
 - (b) Write a for loop that generates 3 normal random variables stored in a vector $\mathbf{y}$ using $\mathbf{z}$ and basic mathematical operations, where the first element of $\mathbf{y}$ has a standard normal distribution and the remaining elements of $\mathbf{y}$ conditional on previous elements have a normal distribution with a mean that of 0.5 times the last element of $\mathbf{y}$ and variance 0.75. Print the code you use to create $\mathbf{z}$ to the rendered `.pdf`.
 - (c) Create a 3×3 matrix $\mathbf{D}$ that creates the same vector $\mathbf{y}$ from $\mathbf{z}$ if you apply $\mathbf{D} \%*\% \mathbf{z}$. You'll need to do some algebra. Print the code you use to create $\mathbf{D}$ to the rendered `.pdf`.
 - (d) Print one line of code you could use to check that your construction of $\mathbf{D}$ in (c) is correct based on what you obtained in (b).

- (e) Using the `arcovariance.R` code, create a 3×3 version of the matrix `C`. You could just take the first 3 rows and first 3 columns of `C`. Print the code you need to compute the Cholesky decomposition of this 3×3 matrix and take its transpose the rendered `.pdf`.
 - (f) Create a plot of the nonzero elements of `D` against the nonzero elements of the transposed Cholesky decomposition in (e). Make sure that your plot is clearly annotated.
 - (g) Write a function that takes n , the length of the vector that you'd like to simulate and ρ , an autocovariance parameter and returns an n dimensional random vector obtained by constructing a covariance matrix C with elements $c_{ij} = \rho^{|i-j|}$, taking its Cholesky decomposition, and multiplying the transpose of its Cholesky decomposition by n independent standard normal random variables (note - order matters). Print the code you use define the function to the rendered `.pdf`.
 - (h) Write a for loop that generates 50 random vectors of length 2 using your function from (a) with $\rho = 0.99$ and stores them in a 2×50 matrix called `mat`. Print the code you use to create this matrix and implement this for loop to the rendered `.pdf`. Include a line of code that ensures that the values you obtain are reproducible by someone else on another computer.
 - (i) Evaluate the sample means, covariances, and correlation of the bivariate random variables you simulated in (b). Summarize your results in a table made using `kable`. Use your knowledge from previous statistics classes to compute these quantities.
 - (j) In at most one sentence, describe how the results in the table in (c) relate to what you would expect to see.
3. Consider a truncated gamma random variable x with shape a and rate b that is constrained to the interval $(0, c)$. The density $g(x|a, b, c)$ of this random variable satisfies:

$$g(x|a, b, c) = \frac{f(x|a, b)}{F(c|a, b)},$$

where $f(x|a, b)$ is the density of a gamma random variable with shape a and rate b and $F(c|a, b)$ is the probability that a gamma random variable with shape a and rate b is less than or equal to c .

- (a) The CDF of this truncated gamma random variable is $G(x|a, b, c) = \frac{F(x|a, b)}{F(c|a, b)}$. Write a function that takes a probability p , Gamma shape and rate parameters a and b , and an upper bound c and returns the inverse CDF of this truncated random variable. Print the code you use define the function to the rendered `.pdf`.
- (b) Use the function from part (a) and uniform random variables to simulate 1,000 draws from a truncated gamma distribution with shape 1 and rate 1 constrained to the interval $(0, 1)$. Make a histogram of the draws using the `hist` function setting the arguments `freq = FALSE` and `br = seq(0, 1, length.out = 10)`.

- (c) Write a for loop that generates 1,000 draws from a gamma distribution with shape 1 and rate 1. What proportion of these draws are less than 1? Provide your code.
- (d) Make two histograms next to each other, one being the histogram you constructed in (b) and the other being a histogram of the draws from (c) that are less than 1, using the same arguments for both. Overlay a line for the density of a truncated gamma random variable with shape 1 and rate 1 constrained to be less than 1 on both plots.
- (e) In at most one sentence, describe whether or not it seems that both methods produce draws from the same distribution.
- (f) In at most one sentence, describe how the number of draws produced by the two methods compares. Is one more efficient than the other?